



**DEPARTAMENTO DE ELETRÓNICA, TELECOMUNICAÇÕES
E INFORMÁTICA**

LICENCIATURA EM ENGENHARIA DE COMPUTADORES E INFORMÁTICA

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**MÉTODOS PROBABILÍSTICOS PARA ENGENHARIA DE
COMPUTADORES E INFORMÁTICA**

PRACTICAL GUIDE No. 0

Task 1- Setup Matlab and Tutorial Course

During this semester the scientific programming language Matlab will be used in the lab classes of MPECI. This task aims to introduce the Matlab language by proposing some very simple problems and by providing some external sources.

- **Matlab instalation:**
<https://www.mathworks.com/academia/tah-portal/universidade-de-aveiro-40766421.html>
[Use as suas credenciais de Utilizador Universal](#)
- Help:
https://www.mathworks.com/support/contact_us.html?s_tid=tah_po_helpbutton_ua.pt
<https://www.ua.pt/pt/stic/matlab>
- Learn MATLAB basics in about two hours:
 Online course: [MATLAB Onramp](https://matlabacademy.mathworks.com/) (<https://matlabacademy.mathworks.com/>)

Task 1- Brief introduction to Matlab

Brief introduction to Matlab based on document “[Matlab num instante](https://sweet.ua.pt/jnvieira/MyDocs/MatlabNumInstante.pdf)” (<https://sweet.ua.pt/jnvieira/MyDocs/MatlabNumInstante.pdf>) created by Professor José Vieira of the University of Aveiro.

Creation of matrices:

The creation of, for example, the following matrix with two rows and three columns

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$$

can be done with the command

```
>> A = [1 2 3; 4 5 6]
```

or, alternatively

```
>> A = [1,2,3;4,5,6]
```

The column and row vectors are special cases of matrices and are created in a similar way. For example, the rows vector $v = [1 \ 2 \ 3]$ is created with

```
>> v = [ 1 2 3]
```

and the corresponding column vector with

```
>> v = [ 1; 2; 3]
```

or transposing the row vector

```
>> v = [1 2 3]'
```

Indexes of matrices:

The element of row i and column j of matrix A is specified as $A(i, j)$. For example, the element of row 1 and column 3 of matrix A is specified as $A(1, 3)$. In Matlab, the element $A(1, 3)$ of the previous matrix A is obtained with

```
>> A(1,3)
```


and the result is

```
ans= 3
```

To change the value of element $A(1, 3)$ to 7, execute

```
>> A(1,3)= 7
```

The indexes of matrices are lists of positive integer values that can be stored in previously created vectors.

For example, we can extract the second row of matrix A either with

```
>> v= A(2,[1 2 3])
```

or creating first a vector with the indexes of the columns

```
>> k= [1 2 3]
```

```
>> v= A(2, k)
```

Operator “:”

Matlab allows the creation of sequences of numbers in a compact way using the operator “:”. The vector

$a = [1, 2, 3, \dots, 10]$ can be created with

```
>> a= 1:10
```

The general notation of the operator “:” is

Initial_number : step : Final_number

and allows the creation of sequences of integer numbers (as in the previous case) or of real numbers.

Some examples are:

```
>> e= 0:pi/20:2*pi
```

```
>> f= 10:-1:-10
```

The operator “:” can be used in the creation of vectors of indexes to obtain compact commands to deal with matrices. For example, the odd columns of matrix

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 4 \\ 6 & 1 & 2 \end{bmatrix} A =$$

can be extracted to a matrix B with

```
>> B=A(1:3,1:2:3)
```

In this case, all rows (of the odd columns) are extracted. For simplified notation, when all rows are to be extracted, we can use

```
>> B=A(:,1:2:3)
```

On the other hand, if the aim is to extract all columns of the first row, this can be done with

```
>> A(1,:)
```


Some problems:

- (a) Create a row vector with a sequence of even numbers starting with number 4 and ending with the number 100.
- (b) Create a row vector with a decreasing sequence of integers starting at 5 and ending at -5.
- (c) Create a row vector with a sequence of equally spaced real numbers with 100 elements belonging to the range $[0 \dots 1]$.
- (d) Create a matrix B with random values using command `>> B= rand(20,30)` (20 rows and 30 columns). Extract to matrix C the sub-matrix of B composed by its rows from 10 to 15 and its columns from 9 to 12.
- (e) Create a sequence, $\underline{x}$, starting with the value $-\pi$, ending with the value π and with a step of $\pi/15$.
- (f) Run the command `>> plot(x, sin(4*pi*x))`. What do you obtain?

